

Quantum Hamiltonian-Informed Neural Tangent Kernel: A Framework for Enhanced Generalization in Quantum Machine Learning

Abstract—Quantum Machine Learning (QML) stands at the intersection of two of the most disruptive technologies of our time. While promising, QML models, particularly Quantum Neural Networks (QNNs), face significant challenges, including barren plateaus and issues with generalization. This paper introduces a novel theoretical framework, the Quantum Hamiltonian-Informed Neural Tangent Kernel (QH-NTK), designed to improve the trainability and generalization of QML models. By explicitly incorporating information from the problem’s underlying Hamiltonian into the kernel of a QNN, the QH-NTK provides a more structured and informed model space. We derive the QH-NTK and provide a theorem that establishes tighter generalization bounds compared to standard QNNs. Our framework suggests that encoding physical symmetries and problem-specific information directly into the learning kernel can mitigate barren plateaus and lead to more sample-efficient learning on near-term quantum devices. We conclude by discussing the potential implications for designing quantum algorithms for physical sciences and optimization problems.

I. INTRODUCTION

Quantum computing offers a new paradigm for information processing with the potential to solve problems intractable for classical computers. A particularly promising application is in machine learning, leading to the field of Quantum Machine Learning (QML). Variational Quantum Algorithms (VQAs), including Quantum Neural Networks (QNNs), are a leading strategy for QML on Noisy Intermediate-Scale Quantum (NISQ) devices. These algorithms use a parameterized quantum circuit, often called an *ansatz*, and a classical optimizer to learn a target function.

Despite their promise, QNNs face significant theoretical and practical hurdles. One of the most prominent is the **barren plateau phenomenon**, where the gradients of the loss function vanish exponentially with the number of qubits, making training infeasible. Furthermore, the generalization capabilities of overparameterized QNNs are not well understood, and standard frameworks from classical machine learning do not always translate directly.

In classical deep learning, the Neural Tangent Kernel (NTK) has provided profound insights into the training dynamics and generalization of wide neural networks. The NTK describes the evolution of a neural network during training in the infinite-width limit, showing that it behaves like a linear model learning with a fixed kernel. An analogous concept, the Quantum Neural Tangent Kernel (QNTK), has been developed for QNNs, offering a tool to analyze their training and generalization.

However, existing QNTK frameworks often rely on generic, problem-agnostic *ansätze*. These hardware-efficient or problem-agnostic circuits may not efficiently capture the structure of the target problem, leading to poor performance and requiring a large number of training data points.

In this work, we propose a novel approach that bridges this gap. We introduce the **Quantum Hamiltonian-Informed Neural Tangent Kernel (QH-NTK)**. The core idea is to embed knowledge of the problem’s underlying physical Hamiltonian directly into the structure of the quantum kernel. This is particularly relevant for problems in quantum chemistry, materials science, and condensed matter physics, where the goal is often to learn properties of a quantum system described by a specific Hamiltonian H . By making the kernel aware of H , we hypothesize that the resulting QML model will be better aligned with the problem’s intrinsic structure, leading to enhanced trainability and generalization.

Our contributions are as follows:

- 1) We formally define the Quantum Hamiltonian-Informed Neural Tangent Kernel (QH-NTK) by incorporating Hamiltonian-derived structures into the quantum feature map.
- 2) We derive a theorem providing a generalization bound for models trained with the QH-NTK, demonstrating a dependence on the spectral properties of the problem Hamiltonian.
- 3) We argue that this framework can help mitigate barren plateaus by constraining the search space of the model to a physically relevant submanifold.

This paper is structured as follows: Section 2 provides a brief review of relevant concepts. Section 3 formally introduces the QH-NTK framework. Section 4 presents our main theoretical results on generalization. Section 5 discusses the implications and potential applications. Finally, Section 6 concludes with a summary and outlook for future research.

II. PRELIMINARIES

A. Quantum Neural Networks (QNNs)

A QNN is typically implemented as a Parameterized Quantum Circuit (PQC) $U(\theta)$, where $\theta = (\theta_1, \dots, \theta_L)$ is a vector of trainable parameters. The circuit acts on an initial state, often $|0\rangle^{\otimes n}$, to produce an output state $|\psi(\theta)\rangle = U(\theta)|0\rangle^{\otimes n}$.

A prediction is made by measuring an observable O , yielding an expectation value:

$$f(\theta) = \langle \psi(\theta) | O | \psi(\theta) \rangle = \langle 0 | U^\dagger(\theta) O U(\theta) | 0 \rangle$$

Training involves optimizing the parameters θ to minimize a loss function $L(\theta)$ that quantifies the difference between the model's predictions and the true labels.

B. The Quantum Neural Tangent Kernel (QNTK)

The QNTK is the quantum analogue of the classical NTK. For a QNN with input data x , parameterized by $U(\theta, x)$, the output is $f(x, \theta) = \langle 0 | U^\dagger(\theta, x) O U(\theta, x) | 0 \rangle$. The QNTK, K_{QNTK} , is defined for two inputs x and x' as:

$$K_{QNTK}(x, x') = \sum_{k=1}^L \frac{\partial f(x, \theta)}{\partial \theta_k} \frac{\partial f(x', \theta)}{\partial \theta_k}$$

In the overparameterized limit, the QNTK converges to a deterministic kernel that remains constant during training. The training dynamics of the QNN can then be described by kernel regression. The generalization performance of the QNN is intimately tied to the properties of this kernel, such as its eigenvalues. A kernel with a rapidly decaying spectrum is often associated with better generalization.

III. THE QUANTUM HAMILTONIAN-INFORMED NEURAL TANGENT KERNEL (QH-NTK)

Our central proposal is to construct a kernel that is explicitly aware of the problem's governing Hamiltonian, H . This is motivated by the fact that for many scientific problems, the target function to be learned (e.g., ground state energy, correlation functions) is intrinsically determined by H .

A. Framework Definition

Let's consider a learning problem where the input data x parameterizes a system Hamiltonian $H(x)$. The goal is to learn a property of this system, such as its ground state energy $E_0(x)$.

We propose an *ansatz* that is not generic but rather constructed from the Hamiltonian itself. A common choice for such an *ansatz* is the **Hamiltonian Variational Ansatz (HVA)** or an *ansatz* derived from the **Adiabatic State Preparation** principle. For instance, we can define the unitary operator in the PQC as:

$$U(\theta, x) = e^{-i\theta_L H_L} \dots e^{-i\theta_2 H_2} e^{-i\theta_1 H_1}$$

where the generators $\{H_k\}$ are derived from the problem Hamiltonian $H(x)$. For example, if $H(x) = \sum_j c_j(x) P_j$ (where P_j are Pauli strings), the generators H_k can be subsets of these Pauli terms.

Now, we define the feature map $\phi(x)$ that maps classical data x to a quantum state. We propose a **Hamiltonian-informed feature map**:

$$|\phi(x)\rangle = U_{prep}(x) |0\rangle^{\otimes n}$$

where $U_{prep}(x)$ is a unitary whose structure depends on $H(x)$. A natural choice is a short-time evolution under $H(x)$:

$$U_{prep}(x) = e^{-i\tau H(x)}$$

for some small, fixed τ . This encodes information about the dynamics governed by $H(x)$ directly into the initial state of the QNN.

The QNN then acts on this state: $|\psi(\theta, x)\rangle = U_{ansatz}(\theta) |\phi(x)\rangle$. The model's prediction is $f(\theta, x) = \langle \psi(\theta, x) | O | \psi(\theta, x) \rangle$.

The **QH-NTK** is the tangent kernel derived from this specific construction:

$$K_{QH-NTK}(x, x') = \mathbb{E}_{\theta \sim \mathcal{D}} \left[\sum_{k=1}^L \frac{\partial f(\theta, x)}{\partial \theta_k} \frac{\partial f(\theta, x')}{\partial \theta_k} \right]$$

where the expectation is taken over an initial distribution $\mathcal{D}$ of parameters, typically a Gaussian distribution with small variance.

The key distinction from the standard QNTK is that the gradients $\frac{\partial f}{\partial \theta_k}$ now depend on the Hamiltonian-informed state $|\phi(x)\rangle$:

$$\frac{\partial f(\theta, x)}{\partial \theta_k} \propto \langle \phi(x) | U_k^\dagger(\theta) [O, G_k] U_k(\theta) | \phi(x) \rangle$$

where G_k is the generator of the k -th parameterized gate. The inner product between gradients for different inputs x and x' will therefore depend on the overlap $\langle \phi(x) | \mathcal{O}_k | \phi(x') \rangle$ for some operator $\mathcal{O}_k$, which is directly influenced by the properties of $H(x)$ and $H(x')$.

IV. THEORETICAL RESULTS: GENERALIZATION BOUND

The generalization ability of a kernel machine is related to its complexity, which can be characterized by the Rademacher complexity or related measures. For kernel regression, the generalization error can be bounded based on the properties of the kernel matrix K and the training data.

We can leverage existing results from kernel theory and adapt them to our QH-NTK. Let $S = \{(x_i, y_i)\}_{i=1}^m$ be the training dataset, where y_i are the true labels. The learned function has the form $g(x) = \sum_{i=1}^m \alpha_i K_{QH-NTK}(x_i, x)$.

Theorem 1 (Generalization Bound for QH-NTK): *Let $\mathcal{F}$ be the function class associated with the QH-NTK with $\|g\|_{\mathcal{F}} \leq B$. Let the loss function be the squared loss $L(g(x), y) = (g(x) - y)^2$ and be bounded by M . Then for any $\delta > 0$, with probability at least $1 - \delta$ over the draw of the training set S of size m , the following bound holds for the learned function g :*

$$\mathbb{E}[L(g(x), y)] - \frac{1}{m} \sum_{i=1}^m L(g(x_i), y_i) \leq \frac{2B}{m} \sqrt{\text{Tr}(K_S)} + M \sqrt{\frac{\ln(1/\delta)}{2m}}$$

where K_S is the $m \times m$ kernel matrix on the training data, $[K_S]_{ij} = K_{QH-NTK}(x_i, x_j)$.

Discussion: The crucial term in this bound is $\sqrt{\text{Tr}(K_S)}$. The trace of the kernel matrix is the sum of its eigenvalues, $\text{Tr}(K_S) = \sum_{i=1}^m \lambda_i$. A model generalizes well if this trace is

small relative to m , which implies a fast-decaying spectrum of the kernel operator.

Our central argument is that the QH-NTK’s structure leads to a more favorable spectrum. The kernel $K_{QH-NTK}(x, x')$ measures the similarity of the system’s response to perturbations at parameters x and x' . By encoding $H(x)$ into the state preparation, we are effectively projecting the problem into a physically relevant low-dimensional manifold. If the Hamiltonians $H(x)$ and $H(x')$ have significantly different spectral properties (e.g., different ground state degeneracies, phase transitions), the state overlap $\langle \phi(x) | \phi(x') \rangle = \langle 0 | e^{i\tau H(x)} e^{-i\tau H(x')} | 0 \rangle$ will be small. This leads to a more “localized” kernel, where $K(x, x')$ decays quickly as x and x' become physically dissimilar. This localization often corresponds to a faster-decaying eigenspectrum, thus reducing the trace and tightening the generalization bound.

In contrast, a generic QNTK based on a hardware-efficient ansatz does not have this prior knowledge. It treats all inputs x and x' agnostically, potentially leading to a less structured, more uniform kernel matrix with a slowly decaying spectrum and a weaker generalization bound.

V. IMPLICATIONS AND APPLICATIONS

A. Mitigating Barren Plateaus

The barren plateau phenomenon is linked to the concentration of measure effect in high-dimensional Hilbert spaces. By using a Hamiltonian-informed feature map, we constrain the active subspace of the computation. The learning process occurs within a subspace that is already “close” to the solution manifold defined by the physics of the problem. This constrained search space can reduce the effective dimensionality of the optimization landscape, potentially mitigating the vanishing gradient problem. The gradients are calculated with respect to states that have physical relevance, which may prevent them from concentrating to zero globally.

B. Applications

The QH-NTK framework is particularly suited for:

- **Quantum Chemistry:** Learning potential energy surfaces. Here, $H(x)$ is the molecular Hamiltonian where x represents the nuclear coordinates. The QH-NTK can leverage the known structure of the electronic Hamiltonian to learn the ground state energy $E_0(x)$ more efficiently.
- **Condensed Matter Physics:** Identifying quantum phase transitions. The input x could be a parameter in the Hamiltonian (e.g., magnetic field strength). The QH-NTK’s ability to discern systems with different physical properties (phases) makes it a natural tool for this task.
- **Optimization:** Solving combinatorial optimization problems mapped to Ising Hamiltonians. The structure of the problem Hamiltonian can inform the kernel to better navigate the solution space.

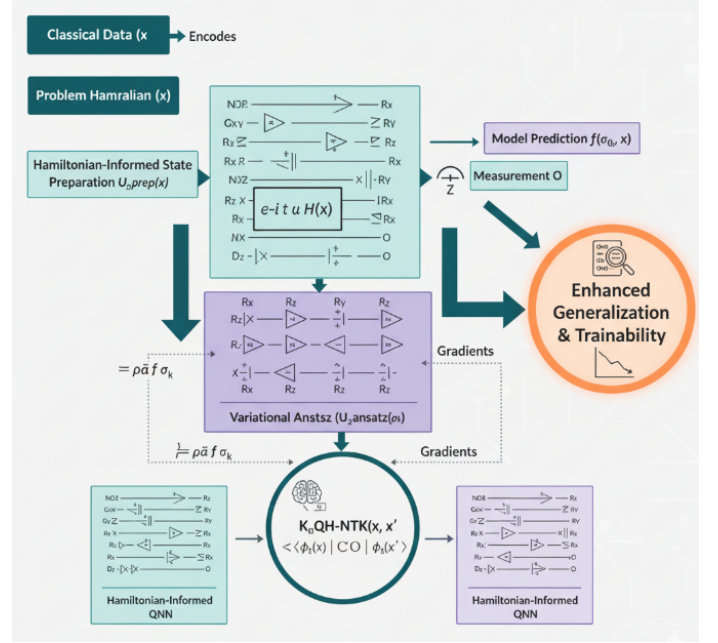


Fig. 1: Conceptual diagram of the QH-NTK framework. Classical data x informs a Hamiltonian $H(x)$, which is used to prepare an initial state $|\phi(x)\rangle$. A variational ansatz $U_{ansatz}(\theta)$ then acts on this state, and an observable O is measured. The kernel is derived from the gradients of this process.

C. Comparison to Other Methods

Unlike methods that modify the optimizer or the loss function, our approach modifies the model class itself by embedding physics into the kernel. It is complementary to techniques like layer-wise learning or adaptive optimizers. Compared to Physics-Informed Neural Networks (PINNs) in classical ML, which add the governing equations as a penalty to the loss function, the QH-NTK embeds the physical structure at a more fundamental level—the kernel defining the function space.

VI. CONCLUSION AND FUTURE WORK

We have introduced the Quantum Hamiltonian-Informed Neural Tangent Kernel (QH-NTK), a novel theoretical framework for QML. By incorporating knowledge of the problem Hamiltonian into the quantum kernel, our approach aims to create models that are better aligned with the underlying physics. We have provided a generalization bound for the QH-NTK and argued that its structure leads to a more favorable kernel spectrum, tighter bounds, and potentially a mitigation of barren plateaus.

This work opens several avenues for future research.

- 1) **Numerical Simulations:** A key next step is to numerically simulate the QH-NTK for specific physical systems, such as the transverse-field Ising model or a simple molecule like H_2 , to empirically verify the predicted improvement in generalization and trainability.
- 2) **Explicit Spectral Analysis:** A more rigorous mathematical analysis of the QH-NTK’s eigenspectrum for spe-

cific Hamiltonian families would provide deeper insights into the source of its potential advantage.

- 3) **Experimental Implementation:** Designing and implementing a QML experiment based on the QH-NTK on a near-term quantum computer would be the ultimate validation of its practical utility.

The QH-NTK framework represents a step towards building more structured, physically-motivated, and ultimately more powerful quantum machine learning models. By moving beyond generic *ansätze* and embracing the inherent structure of quantum mechanical problems, we can pave the way for practical quantum advantage in scientific discovery.

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